the −5/3 slope indicated by the inserted broken line. While not claiming any physical similarity between the separated flow behavior described here and the case of the homogeneous turbulent flow, it is worthy of noting that the -5/3 slope is predicted by the theory of universal equilibrium in homogeneous turbulent flow (see Reference 34, "The Structure of Turbulent Shear Flow," by A. A. Townsend). The time span during which the spectral data of Figure 14 were acquired was t = 363.4 - 364.22. During this time, the flow on the complete upper wing surface became separated. A typical power spectrum plot for an inboard pressure station located in the trailing edge flap, Pressure Station No. 18, Figure 3, acquired at the same time interval is shown in Figure 15. Figure 15 shows that the PSD level was quite high at this time. It represented an overall peak in spectral power for Station No. 18 during the complete maneuver. A major cause of the high spectral power at Station No. 18 was contributed to the high turbulence when the local flow became separated.

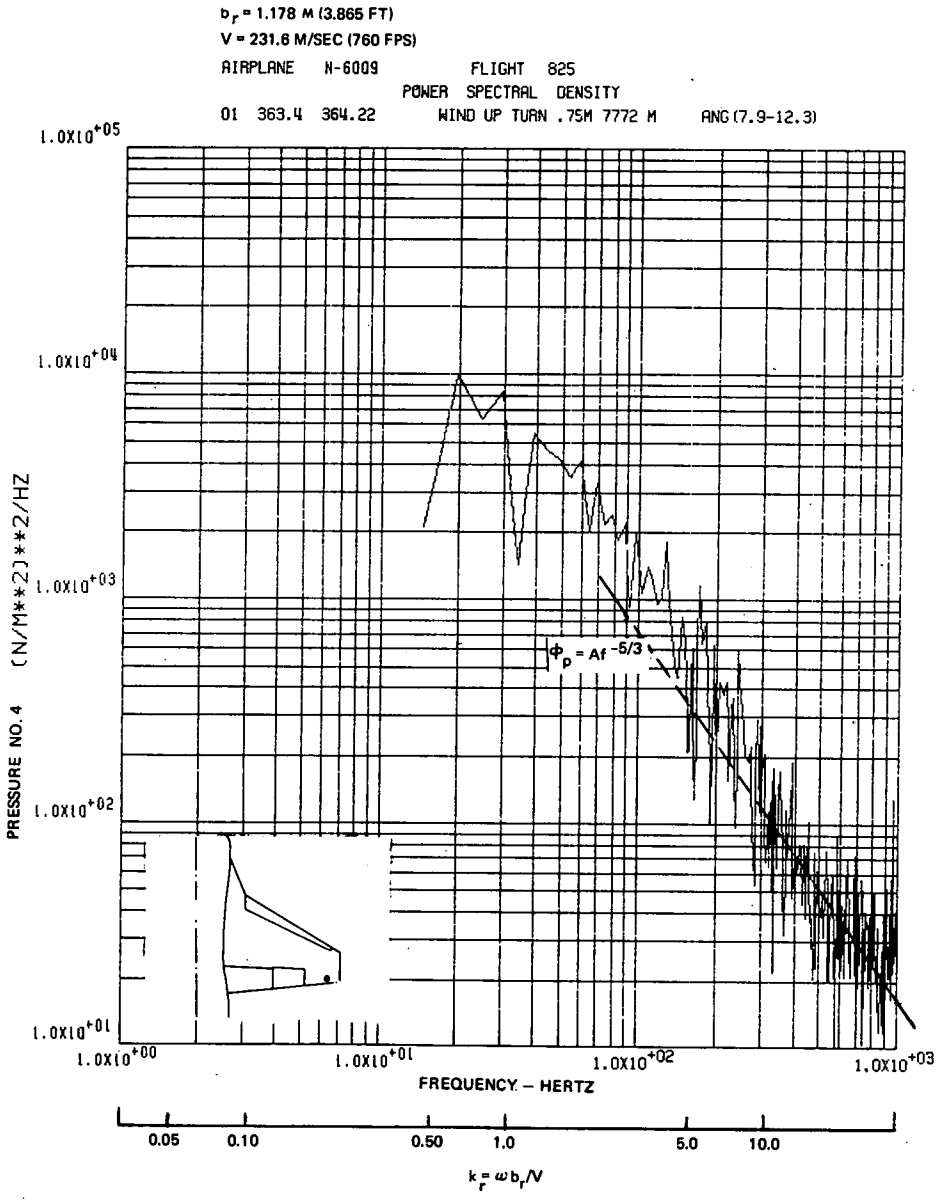

FIGURE 14. POWER SPECTRUM OF BUFFET PRESSURE AT WING STATION 128.31, 90% CHORD POSITION, FOR M_o = 0.75, h = 7,772 m, δ_n = 0°, δ_f = 0°